



Lab 08: Inter-VLAN Routing

Router1 (2911) · Switch (2960) · PC0 · PC1 · PC2 · PC3

Lab #08

Topic: Inter-VLAN Routing

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Add Router1 (2911) to enable routing between VLANs from Lab 07.
- Configure trunk link from switch Gig0/1 to Router1 Gig0/0.
- Set up 802.1Q sub-interfaces on Router1 — one per VLAN.
- Assign PC0–PC3 to four separate VLANs on the switch.
- Verify full inter-VLAN communication between all four PCs.

2. Network Topology

Router1 (2911) on the left connects via Gig0/0 to the 2960-24TT switch on Gig0/1 (trunk link). PC0 on Fa0/1, PC1 on Fa0/2, PC2 on Fa0/3, PC3 on Fa0/4 — each on a separate VLAN. Router1 sub-interfaces act as the gateway for each VLAN.

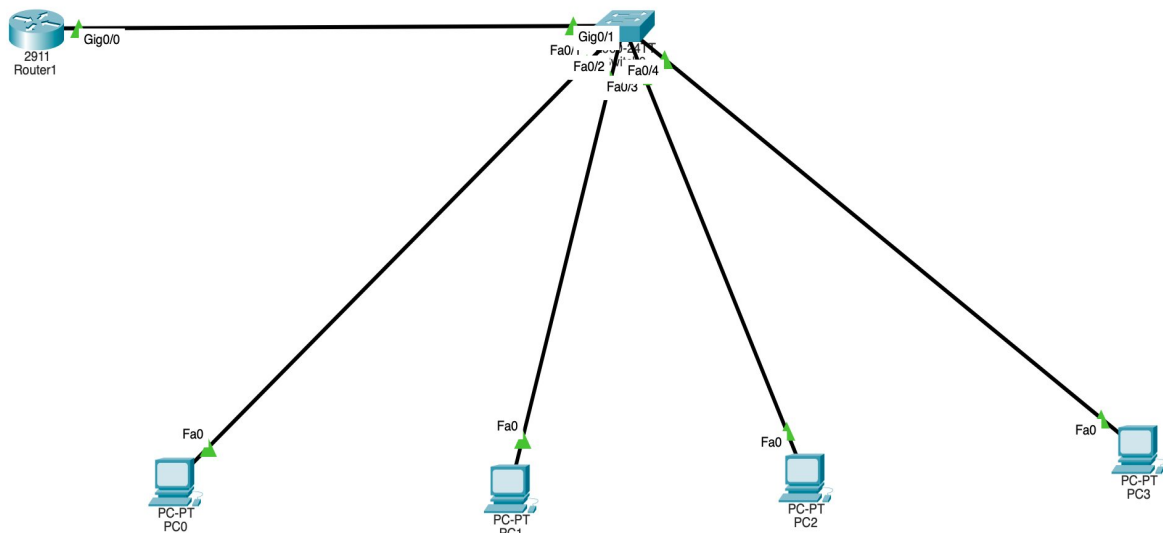


Figure 1 – Router1 (2911) trunk to Switch0, PC0–PC3 on VLANs 10/20/30/40

Device	Switch Port	VLAN	IP Address	Gateway
PC0	Fa0/1	VLAN 10	192.168.10.10	192.168.10.1
PC1	Fa0/2	VLAN 20	192.168.20.10	192.168.20.1



PC2	Fa0/3	VLAN 30	192.168.30.10	192.168.30.1
PC3	Fa0/4	VLAN 40	192.168.40.10	192.168.40.1

3. Configuration

3.1 Switch – VLANs, Access Ports, Trunk

```
Switch(config)#vlan 10
Switch(config-vlan)#name VLAN10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name VLAN30
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name VLAN40
Switch(config-vlan)#exit
Switch(config)#interface Fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface Fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface Fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface Fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit
Switch(config)#interface GigabitEthernet0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch#write memory
```

3.2 Router1 – Sub-interfaces



```
Router1(config)#interface GigabitEthernet0/0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#interface GigabitEthernet0/0.10
Router1(config-subif)#encapsulation dot1Q 10
Router1(config-subif)#ip address 192.168.10.1 255.255.255.0
Router1(config-subif)#exit
Router1(config)#interface GigabitEthernet0/0.20
Router1(config-subif)#encapsulation dot1Q 20
Router1(config-subif)#ip address 192.168.20.1 255.255.255.0
Router1(config-subif)#exit
Router1(config)#interface GigabitEthernet0/0.30
Router1(config-subif)#encapsulation dot1Q 30
Router1(config-subif)#ip address 192.168.30.1 255.255.255.0
Router1(config-subif)#exit
Router1(config)#interface GigabitEthernet0/0.40
Router1(config-subif)#encapsulation dot1Q 40
Router1(config-subif)#ip address 192.168.40.1 255.255.255.0
Router1(config-subif)#exit
Router1#write memory
```

4. Verification

```
PC0>ping 192.168.20.10
PC0>ping 192.168.30.10
PC0>ping 192.168.40.10
Router1#show ip route
```

5. Key Takeaways

- Router-on-a-Stick routes between all VLANs using ONE physical trunk link.
- The parent interface must be 'no shutdown' — sub-interfaces inherit its state.
- encapsulation dot1Q tags outgoing frames with the correct VLAN ID.
- Four VLANs = four sub-interfaces = four /24 subnets on Router1.
- A Layer 3 switch does this in hardware — faster and more scalable than a router.